
SUMMARY

Healthcare data analyst and Computer Science graduate with experience building SQL and Python-based data workflows, PostgreSQL-backed systems, validation checks, reporting-ready outputs, and reproducible analytics pipelines. Strong foundation in relational data modeling, joins, aggregations, data quality review, analytical documentation, trend analysis, Excel, Power BI, Python, SQL, pandas, NumPy, Matplotlib, FastAPI, SQLAlchemy, PostgreSQL, and Docker. Current Beth Israel Deaconess Medical Center Patient Care Technician in hematology/oncology with hands-on Epic EHR documentation experience and familiarity with clinical data accuracy, patient status tracking, and healthcare workflows.

EDUCATION

University of Massachusetts Amherst | College of Information and Computer Sciences

Bachelor of Science in Computer Science, May 2025 | GPA 3.408 (Spring 2025 term GPA 3.936)

Relevant coursework: Introduction to Algorithms, Operating Systems, Computer Networks, Software Engineering, Statistics, Machine Learning, Linear Algebra, Computer Systems Principles, Programming with Data Structures, Reasoning Under Uncertainty

PROJECTS

Clinical Document RAG Assistant | Python, FastAPI, PostgreSQL, pgvector, SQLAlchemy, OpenAI API, Docker

- Built a backend document-ingestion and retrieval pipeline for PDF-based clinical-style documents, supporting file upload, text extraction, overlapping chunking, OpenAI embedding generation, PostgreSQL/pgvector storage, semantic search, and source-grounded answer generation.
- Designed SQLAlchemy models for uploaded documents and document chunks with one-to-many relationships, persistent metadata, chunk content, and vector embeddings so retrieved answers can be traced back to source PDFs.
- Implemented modular FastAPI services and endpoints for document ingestion, semantic search, and question answering, returning source previews through documented OpenAPI/Swagger interfaces.

Longitudinal Lesion Tracking REST API | Python, FastAPI, SQLAlchemy, PostgreSQL, Docker Compose

- Built a Dockerized PostgreSQL-backed REST API with 26 endpoints supporting CRUD operations, nested read views, and analytics-style queries for structured longitudinal measurement records.
- Designed a normalized relational schema with foreign keys, cascading deletes, scoped uniqueness constraints, and validation checks to prevent duplicate or inconsistent records.
- Created FastAPI/OpenAPI documentation and automated demo data workflows to support repeatable testing, endpoint validation, and handoffs.

Song Recommender API | Python, FastAPI, SQLAlchemy, PostgreSQL, scikit-learn, Docker

- Built a catalog-backed recommendation API supporting Search → select up to 20 seed tracks → return up to 20 similarity-based recommendations through stable request/response schemas.
- Created an offline indexing pipeline using StandardScaler, optional PCA embeddings, and cosine NearestNeighbors retrieval, persisting fitted artifacts plus feature-order metadata to keep recommendations reproducible across runs without retraining.
- Focused on consistent outputs and predictable data contracts to support downstream analysis and troubleshooting when inputs drift or data quality issues appear.

Custom ML Benchmarking Suite | Python, NumPy, pandas, Matplotlib, scikit-learn

- Built a benchmarking framework that standardizes preprocessing and evaluation using normalization and k-fold cross-validation to avoid one-off results.
- Summarized accuracy and F1 with clean tables and Matplotlib figures suitable for reporting and technical write-ups.
- Best results included accuracy = 98.55% for k-nearest neighbors, accuracy = 93.05% for random forest, accuracy = 92.8% for the neural network.

SKILLS

Healthcare Data: Epic EHR documentation workflows, clinical data monitoring, patient status tracking, healthcare documentation

Languages: Python, SQL, JavaScript, TypeScript, Java, C, C++

Data: PostgreSQL, relational modeling, joins/aggregations, constraints, validation rules, structured CSV-style workflows

Engineering: FastAPI, SQLAlchemy, REST APIs, OpenAPI/Swagger, Docker, Docker Compose, Git/GitHub

Analytics and Reporting: Excel, Power BI, reporting-ready outputs, dashboards, trend analysis, data validation, data quality checks

EXPERIENCE

Patient Care Technician (Boston, MA) | Beth Israel Deaconess Medical Center

April 2026 - Present

- Monitor, document, and validate patient clinical data in Epic EHR workflows to support accurate patient status tracking, continuity of care, and escalation decisions.
- Review patient vitals and condition trends in a high-acuity hematology/oncology environment, identifying abnormal changes and communicating findings to nursing staff.
- Maintain accurate healthcare documentation while coordinating with multidisciplinary care teams in a fast-paced clinical environment, building familiarity with Epic workflows, documentation quality, and clinical data accuracy.

IT Support Administrator (Boston, MA) | Pope John Paul II Polish Language School

September 2017 - August 2023

- Provided onsite technical support for classroom and administrative technology, troubleshooting computers, Wi-Fi connectivity, display systems, printers, AV equipment, and common software issues during live weekend school operations.